

SIMATS ENGINEERING
ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I P.Raghavendra Reddy/1911260258 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.RAGHAVENDRA REDDY

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

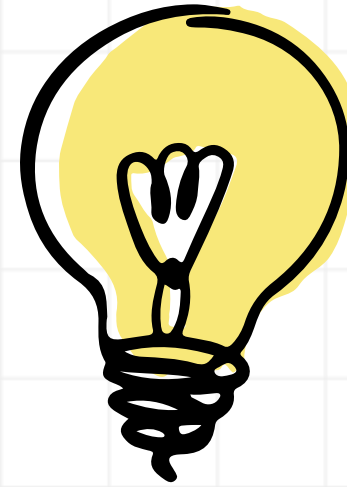
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

DEFINITION

CLOUD COMPUTING IS A MODEL THAT PROVIDES SHARED, CONFIGURABLE COMPUTING RESOURCES DELIVERED OVER A NETWORK WITH MINIMAL MANAGEMENT EFFORT. THESE RESOURCES INCLUDE INFRASTRUCTURE AS A SERVICE (IAAS), PLATFORM AS A SERVICE (PAAS), AND SOFTWARE AS A SERVICE (SAAS), EACH OFFERING DIFFERENT LEVELS OF CONTROL AND ABSTRACTION.



BENEFITS

CLOUD COMPUTING OFFERS SEVERAL ADVANTAGES THAT MAKE IT ATTRACTIVE FOR BUSINESSES AND INDIVIDUALS:

- COST EFFICIENCY** — ELIMINATES THE NEED FOR UPFRONT HARDWARE PURCHASES AND REDUCES MAINTENANCE COSTS.
- SCALABILITY** — AUTOMATICALLY ADJUSTS RESOURCES TO HANDLE VARYING WORKLOADS, IDEAL FOR GROWING BUSINESSES OR UNPREDICTABLE DEMAND.
- FLEXIBILITY AND MOBILITY** — USERS CAN ACCESS APPLICATIONS AND DATA FROM ANYWHERE WITH AN INTERNET CONNECTION.

CHARACTERISTICS

THESE CHARACTERISTICS DISTINGUISH CLOUD COMPUTING FROM TRADITIONAL ON-PREMISES COMPUTING:

- ON-DEMAND SELF-SERVICE** — USERS CAN PROVISION COMPUTING RESOURCES AUTOMATICALLY WITHOUT HUMAN INTERVENTION FROM THE PROVIDER.
- BROAD NETWORK ACCESS** — SERVICES ARE ACCESSIBLE OVER THE INTERNET FROM VARIOUS DEVICES (LAPTOPS, PHONES, TABLETS).
- RESOURCE POOLING** — PROVIDERS USE MULTI-TENANT ARCHITECTURES, POOLING RESOURCES TO SERVE MULTIPLE CUSTOMERS EFFICIENTLY.

CLOUD COMPUTING

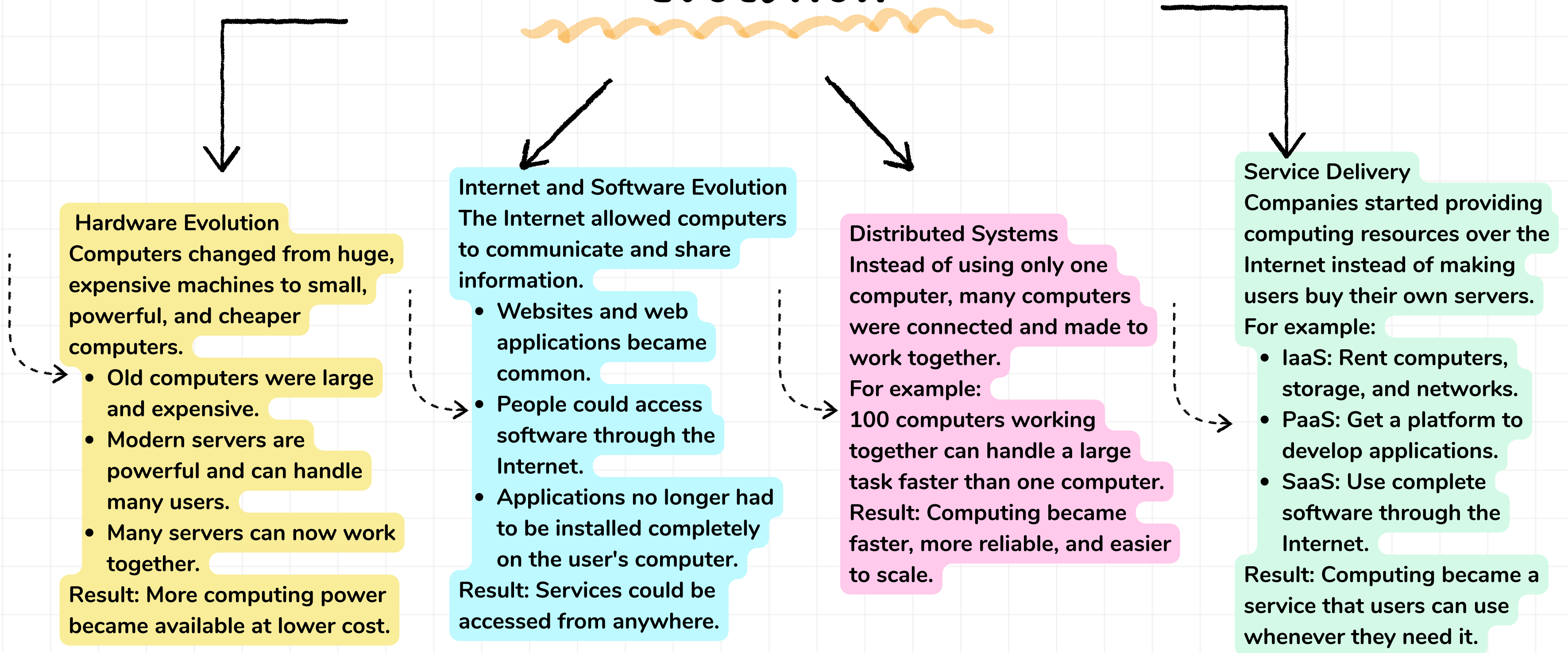
LIMITATIONS

DESPITE ITS ADVANTAGES, CLOUD COMPUTING HAS CONSTRAINTS THAT ORGANIZATIONS MUST CONSIDER:

- SECURITY AND PRIVACY CONCERNS** — STORING DATA ON THIRD-PARTY SERVERS INTRODUCES RISKS RELATED TO UNAUTHORIZED ACCESS, COMPLIANCE, AND DATA SOVEREIGNTY.
- DEPENDENCY ON INTERNET CONNECTIVITY** — CLOUD SERVICES REQUIRE STABLE, HIGH-SPEED INTERNET; OUTAGES CAN DISRUPT OPERATIONS.
- LIMITED CONTROL** — USERS HAVE LESS CONTROL OVER INFRASTRUCTURE, CONFIGURATIONS, AND UPDATES COMPARED TO ON-PREMISES SYSTEMS.

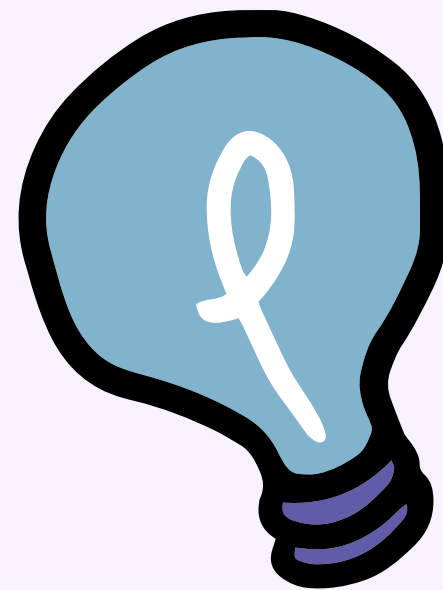
EVOLUTION OF CLOUD COMPUTING

EVOLUTION



SERVICE MODELS

SAAS – SOFTWARE AS A SERVICE
PROVIDER → INFRASTRUCTURE, OS,
SOFTWARE, MAINTENANCE
USER → DATA, SETTINGS, USAGE
EXAMPLE → GMAIL, GOOGLE DOCS



IaaS – Infrastructure as a Service
Provider → Servers, storage,
networking, virtualization
User → Operating system,
applications, data
Example → Virtual machine

XAAS – ANYTHING AS A SERVICE
PROVIDER → PROVIDES THE REQUIRED
TECHNOLOGY/SERVICE
USER → USES AND CONFIGURES THE SERVICE
EXAMPLES → STORAGE-AS-A-SERVICE, DATABASE-
AS-A-SERVICE

🔑 MAIN RELATIONSHIP

PaaS – Platform as a Service
Provider → Infrastructure, OS, runtime,
development platform
User → Application, data
Example → App development platform

VIRTUALIZATION AND WEB SERVICES AS ENABLERS OF CLOUD

1. Virtualization

- **Meaning** → Creates virtual versions of physical resources
- **Virtual Machines** → One physical server can run many VMs
- **Resource Sharing** → CPU, memory, storage shared efficiently
- **Isolation** → Different users/apps can run separately
- **Scalability** → New virtual machines can be created quickly
- **Cloud Deployment** → Makes it easy to provide computing resources on demand

Simple flow:

Physical Server → Virtual Machines → Resource Sharing → Fast Deployment → Scalability

Web Services

- **Meaning** → Software services communicate through the Internet
- **APIs** → Allow applications to communicate
- **Remote Access** → Services can be accessed from anywhere
- **Service Integration** → Different applications can work together
- **Automation** → Resources and services can be requested automatically
- **Scalability** → Services can handle increasing users and requests

Simple flow:

Internet → Web Services/APIs → Application Communication → On-Demand Access → Scalability

Connection to Cloud Virtualization

- Efficiently uses physical resources
- Creates flexible virtual resources

Supports cloud deployment & scalability

Web Services

- Connects applications and users
- Provides services over the Internet

Supports cloud access & scalability

Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

- CONCEPT MAP-1
- The concept map is effective because it clearly connects the main principles of cloud computing and shows how they work together.
- Service models: IaaS, PaaS, SaaS, and XaaS show different levels of control and service provided by the cloud.
- APIs: APIs allow applications and cloud services to communicate, integrate, and automate tasks.
- Elasticity: Cloud resources can be increased or decreased according to demand.
- Scalability: Virtualization and distributed systems help cloud services support more users and larger workloads.
- Virtualization: It allows multiple virtual machines to share physical resources, improving utilization and making deployment faster.
- Responsibility boundaries: The map shows that the provider manages more resources as we move from IaaS to PaaS and SaaS, while the user's responsibility decreases.
- Overall, the strongest design decision is connecting these concepts instead of showing them separately. This makes it easier to understand how cloud services are delivered, scaled, managed, and accessed.

Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

- CONCEPT MAP-2
- The concept map shows that scalability, elasticity, and resource management are important for an effective cloud architecture.
- Scalability: Allows the system to handle more users and workloads by adding resources.
- Elasticity: Allows resources to automatically increase or decrease based on demand.
- Resource management: Ensures CPU, memory, storage, and network resources are used efficiently.
- Virtualization: Supports resource sharing and allows virtual machines to be created quickly.
- Managed compute: Reduces the user's responsibility for managing servers and infrastructure.
- APIs: Connect cloud services and support automation and resource management.
- Service models: IaaS, PaaS, and SaaS determine how much infrastructure the provider manages and how much control the user has.
- If scalability and elasticity were absent, the system could become slow or unavailable during high demand, while unused resources could increase costs during low demand. Without effective resource management, resources may be wasted, causing poor performance and higher costs. Therefore, these capabilities are essential for making cloud systems flexible, reliable, and cost-effective.

Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

- CONCEPT MAP-3
- The APIs, web services, and cloud services in the concept map are selected because they support communication, scalability, flexibility, and efficient resource management.
- APIs: Allow different applications and cloud services to communicate and exchange data. They also support automation and integration.
- Web services: Provide applications and resources through the Internet, making services accessible from different devices and locations.
- Cloud services: IaaS, PaaS, and SaaS provide different levels of control and management. The suitable model can be selected based on the user's requirements.
- Virtualization: Allows multiple virtual machines to share physical resources, improving resource utilization and reducing infrastructure costs.
- Elasticity and scalability: Cloud resources can increase or decrease according to workload, helping maintain performance and avoid unnecessary resource usage.
- Responsibility boundaries: IaaS gives users more responsibility, while PaaS and SaaS shift more management to the provider.
- Overall, these components make the system flexible, scalable, efficient, and easier to manage while allowing users to choose the appropriate level of control and service.

Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

CONCEPT MAP-4

1. The components in the concept map are closely connected and together improve the reliability, performance, and user experience of cloud systems.
2. Service models: IaaS, PaaS, and SaaS define different levels of control and responsibility, helping users select the service that best fits their needs.
3. APIs and web services: APIs connect applications and cloud services, enabling communication, integration, and automation.
4. Virtualization: Allows resources to be shared efficiently and enables quick deployment of virtual machines or managed compute resources.
5. Scalability and elasticity: These work with virtualization to add or remove resources according to workload. This maintains performance during high demand and reduces wasted resources during low demand.
6. Resource management: Efficient allocation of CPU, memory, storage, and networking improves system performance and reduces costs.
7. Responsibility boundaries: Moving from IaaS to PaaS and SaaS shifts more management to the provider, reducing the user's maintenance effort.
8. Overall, these relationships create a flexible and reliable architecture. APIs improve connectivity, virtualization improves resource use, elasticity and scalability maintain performance, and appropriate service models reduce management effort. Together, they provide a better user experience, availability, and efficiency.

Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Creating the concept map helped me understand that cloud computing is not just about storing data online, but a combination of hardware, virtualization, distributed systems, APIs, web services, and service models working together.

While developing the map, I understood how virtualization supports resource sharing, while APIs and web services enable communication between applications. I also understood that scalability and elasticity help cloud systems handle changing workloads. The service models, such as IaaS, PaaS, and SaaS, helped me understand the different responsibilities of cloud providers and users.

If I redesigned the map, I would make the relationships between virtualization, APIs, scalability, elasticity, and service models more visible. I would also use clearer arrows and fewer words to make the map easier to read and understand. Finally, I would add a few real-world examples to connect the concepts with practical cloud services.